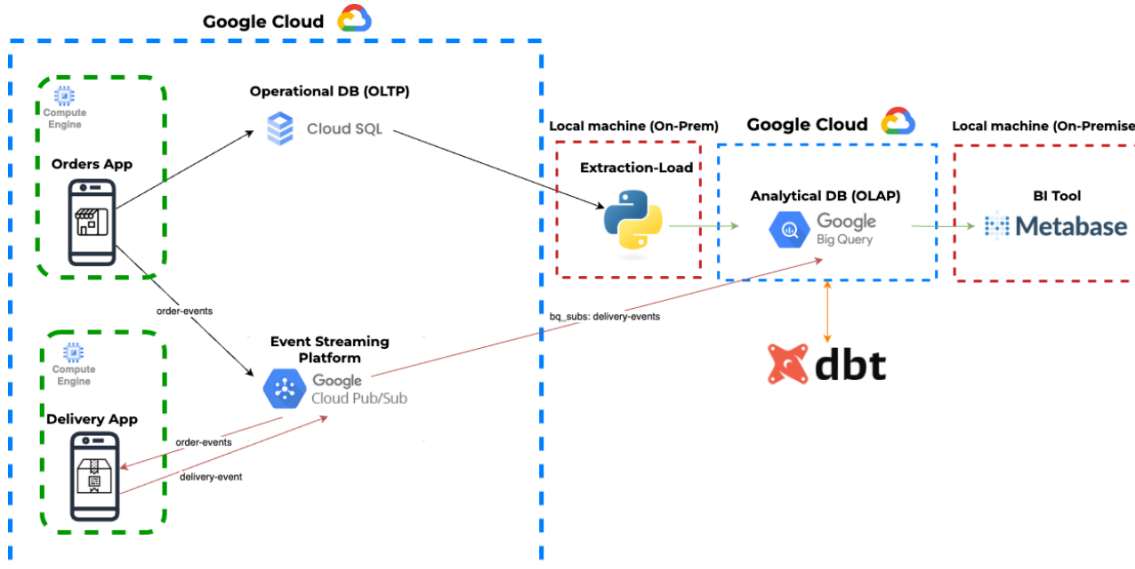


GOOGLE CLOUD



1. Base image that will be used for virtual machines. It's necessary to install Docker and clone the git repository.

Overview

Security risk overview

Virtual machines

- VM instances
- Instance templates
- Sole-tenant nodes
- Machine images
- TPUs
- Committed-use discounts
- Reservations
- Migrate to Virtual Machines

A machine image contains a VM's properties, metadata, permissions and data from all its attached disks. You can use a machine image to create, backup or restore a VM. [Learn more](#)

Filter Enter property name or value

Status	Name	Source instance	Machine type	Architecture	Storage location	Creation time	Actions
☒	base-image:entregable	orders-app	e2-micro	x86_64	eu	Mar 16, 2026, 10:16:42 am UTC+01:00	

2. Create Pub/Sub topics.

Overview

Topics

- Subscriptions
- Snapshots
- Schemas
- Pub/Sub Lite
- Lite reservations

List Metrics

Filter Filter topics

Topic ID	Encryption key	Topic name	Retention	Ingestion source
☒	Google-managed	projects/entregablegcp/topics/delivery-events	—	—
☒	Google-managed	projects/entregablegcp/topics/order-events	—	—

3. Create the instance for orders-app and delivery-app

Overview

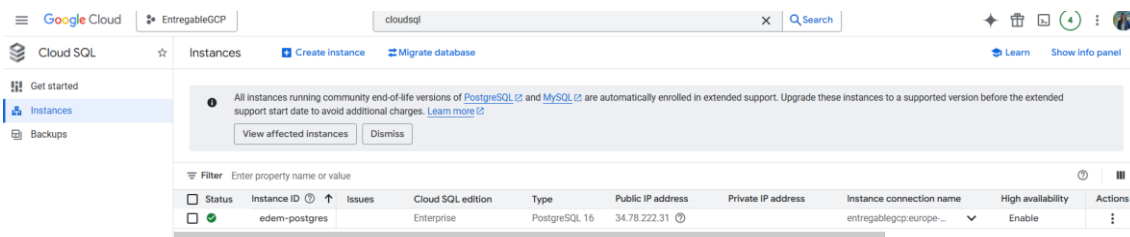
VM instances

- Instance templates
- Sole-tenant nodes
- Machine images
- TPUs

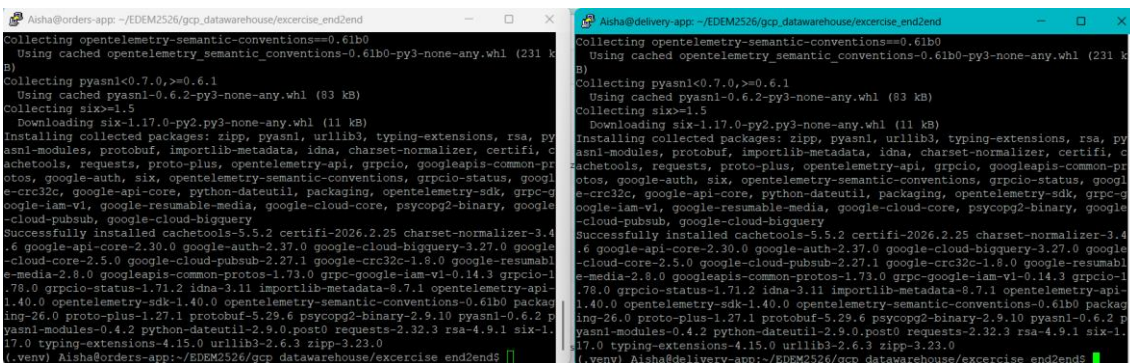
Filter Enter property name or value

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☒	delivery-app	europe-west1-b			10.132.0.4 (nic0)	34.78.160.21 (nic0)	SSH
☒	orders-app	europe-west1-b			10.132.0.3 (nic0)	34.62.197.113 (nic0)	SSH

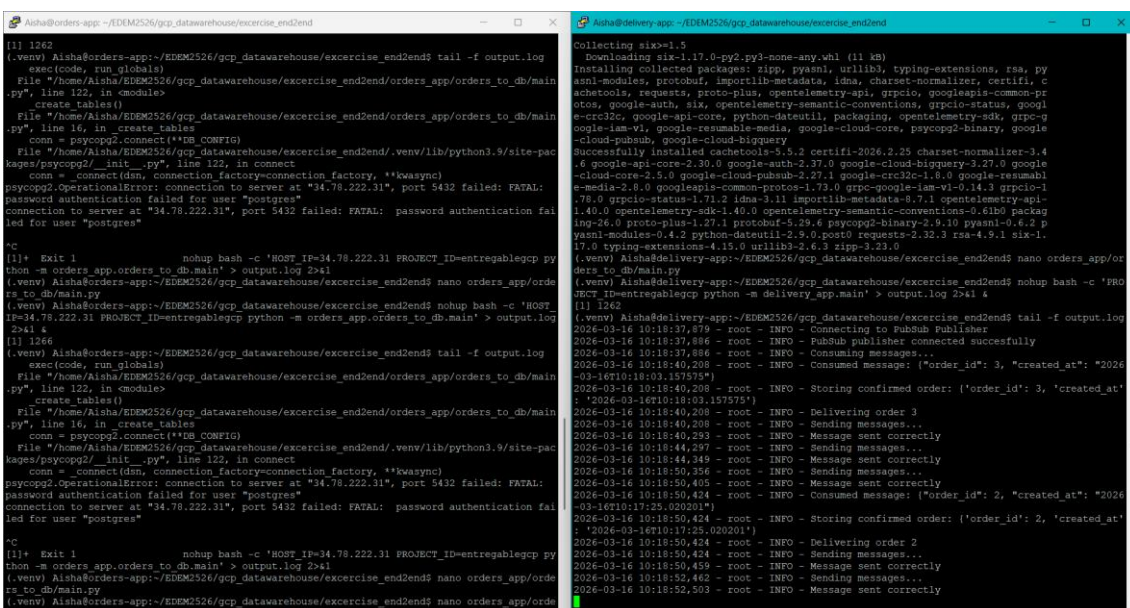
4. Instance CloudSQL. Create the user and the database.



5. Install the requirements in both instances.



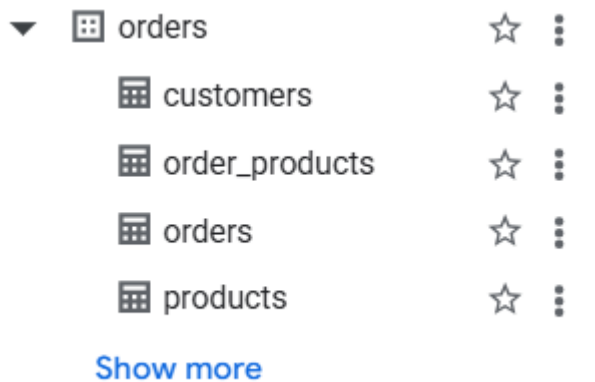
6. Start running both instances.



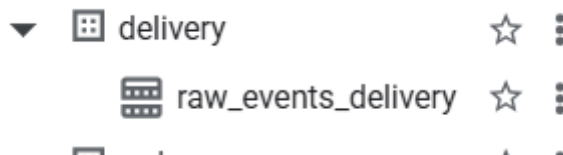
Problem: Change the password in main.py from orders-app. Used nano, modified the script (change the password) and then it worked.

7. BigQuery

Orders-app:



Delivery-app:



Create subscription to write in BigQuery.

8. Sincronize Postgress to BigQuery

set POSTGRES_IP=34.78.222.31

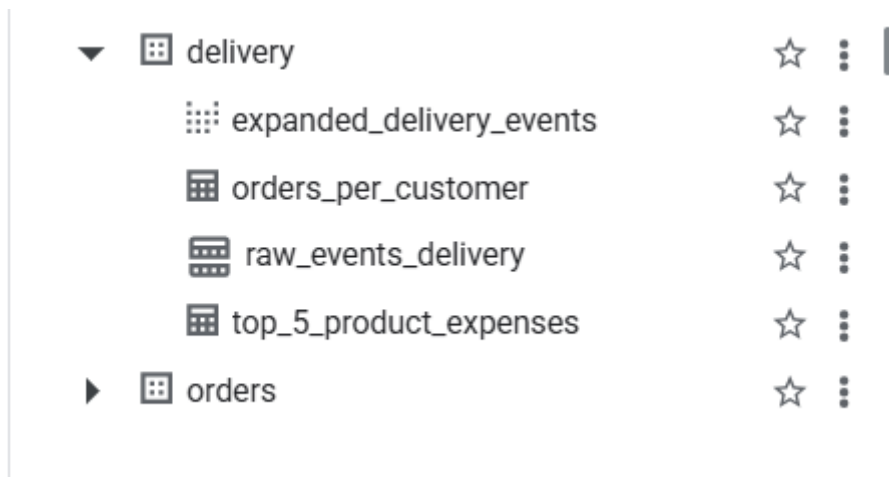
set GCP_PROJECT=entregablegcp

python -m analytical_layer.el_orders.main

```
(.venv) C:\Users\Aisha\EDM2526\gcp_datawarehouse\exercice_end2end>python -m analytical_layer.el_orders.main
2026-03-16 12:07:08,665 - root - INFO - Starting synchronization.
2026-03-16 12:07:08,666 - root - INFO - Fetching data from OLTP for customers.
2026-03-16 12:07:08,933 - root - INFO - Connected to PostgreSQL successfully.
2026-03-16 12:07:09,005 - root - INFO - Closed PostgreSQL connection.
2026-03-16 12:07:11,893 - root - INFO - Connected to BigQuery successfully.
2026-03-16 12:07:12,363 - root - INFO - Fetched schema for entregablegcp.orders.customers.
2026-03-16 12:07:12,363 - root - INFO - Converting data for customers.
2026-03-16 12:07:14,753 - root - INFO - Connected to BigQuery successfully.
2026-03-16 12:07:20,199 - root - INFO - Inserted 10 records into entregablegcp.orders.customers.
2026-03-16 12:07:20,200 - root - INFO - Successfully wrote 10 records to BigQuery table customers.
2026-03-16 12:07:20,201 - root - INFO - Fetching data from OLTP for orders.
2026-03-16 12:07:20,436 - root - INFO - Connected to PostgreSQL successfully.
```

Problem: Change the password in main.py.

9. DBT for delivery events.



Problems: BigQuery location mismatch by updating the region in the profiles.yml file.

Create service account to connect to BigQuery. Manage keys and obtain JSON.

← Create service account

✓ Create service account

2 Permissions (optional)

Grant this service account access to EntregableGCP so that it has permission to complete specific actions on the resources in your project.
[Learn more](#)

Role

BigQuery Data Viewer

Access to view datasets and all of their contents

IAM condition (optional) ⓘ

+ Add IAM condition

+

Role

BigQuery Job User

Access to run jobs

IAM condition (optional) ⓘ

+ Add IAM condition

+

+ Add another role

Continue

3 Principals with access (optional)

Done Cancel

Visualize the data

